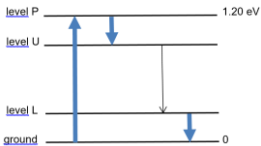


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	More electrons occupy Level U than Level L Accept more electrons in higher state than lower state.	1			1		
		(ii)	All 3 arrows correct. 		1		1		
		(iii)	Greater probability of stimulated emission [of photon] ... (1) ...than absorption [of photon] (1) So extra <u>photon</u> produced (1)	3			3		
	(b)		Calculation of $1.92 \times 10^{-19} \text{ J}$ (1) Use of $E = hf$ or $E = \frac{hc}{\lambda}$ (1) $f < 2.9 \times 10^{14} \text{ Hz}$ ecf or $\lambda > 1.0 \times 10^{-6} \text{ m}$ ecf so visible light not produced (1) Alternative: Use of $\frac{hc}{e\lambda}$ (1) eV values calculated at 400 nm and 700 nm i.e. 3.1 eV and 1.78 eV (1) Lasing transition less than 1.20 eV so visible light not produced (1) Alternative: Lasing transition $< 1.20 \text{ eV}$ (1) Red photon is $\approx 2 \text{ eV}$ (1) Therefore infra-red photon produced (1)			3	3	2	
			Question 7 total	4	1	3	8	2	0